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Digital Image Processing

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CLUSTERING AND SUPERPIXELS

K-Means Clustering

- ▶ Clustering is one of classical approaches of region segmentation
- ▶ The basic idea is to partition a set of observations, Q , into a specified number of clusters, k .
- ▶ Each observation is assigned to the cluster with the nearest mean. Each mean is called the ***prototype*** of its cluster.
- ▶ A k-means algorithm is an iterative procedure that refines the means until convergence is achieved

K-Means Clustering

- ▶ Let $\{z_1, z_2, \dots, z_Q\}$ be the set of vector observations. Each vector has the form

$$z = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix}$$

- ▶ Components of vector z represent numerical pixel attributes. Therefore, depending on the type of image, grayscale images only have one scalar intensity value in each pixel, while in RGB colour images are 3-D vector.

K-Means Clustering

- ▶ K-means clustering is to partition set Q into k disjoint cluster sets $C = \{C_1, C_2, \dots, C_k\}$ in which this condition is satisfied

$$\arg \min_c \left(\sum_{i=1}^k \sum_{z \in C_i} \|z - \mathbf{m}_i\|^2 \right)$$

- ▶ $\mathbf{m}_i$ is the mean vector (called centroid) of set C_i and $\|z - \mathbf{m}_i\|$ is the Euclidean distance from a sample in C_i to the mean $\mathbf{m}_i$
- ▶ Therefore, this equation is to find the sets $C = \{C_1, C_2, \dots, C_k\}$ such that sum of these distances is minimum

K-Means Clustering

- ▷ However, finding the minimum is an NP-hard problem. Therefore, there are no practical solutions. Only attempts on a number of heuristic methods to find approximations have been proposed.
- ▷ The algorithm which is considered to be the standard k-means is as follow
 1. Specify an initial set of means $\mathbf{m}_i$
 2. Assign samples to the set whose the mean is the closest
$$\mathbf{z}_q \rightarrow C_i \text{ if } \|\mathbf{z}_q - \mathbf{m}_i\|^2 < \|\mathbf{z}_q - \mathbf{m}_j\|^2 \quad j = 1, 2, \dots, k (j \neq i); q = 1, 2, \dots, Q$$
 3. Update the cluster means
$$\mathbf{m}_i = \frac{1}{|C_i|} \sum_{\mathbf{z} \in C_i} \mathbf{z} \quad i = 1, 2, \dots, k$$
 4. Test if the means of the previous steps and the current one are different less than a threshold. If not, repeat step 2

K-Means Clustering



Original image

k = 3

Superpixels

- ▶ Superpixels is an approach where replacing the standard pixel grid by grouping pixels into primitive regions that are more perceptually meaningful
- ▶ The objective are to lessen computational load and improve the performance of segmentation algorithms

Superpixels

- ▶ Superpixels can be generated using an algorithm called SIMPLE LINEAR ITERATIVE CLUSTERING (SLIC)
- ▶ This algorithm modifies k-means vectors into more dimensional vectors. For example, a vector in RGB colour system is 5-dimensional consists of three-colour components and its spatial coordinates

$$\mathbf{z} = \begin{bmatrix} r \\ g \\ b \\ x \\ y \end{bmatrix}$$

Superpixels

► The SLIC superpixel algorithm is as follow

1. Compute the initial superpixel cluster centers

$$\mathbf{m}_i = [r_i \ g_i \ b_i \ x_i \ y_i]^T, i = 1, 2, \dots, n_{sp}$$

2. For each cluster center, compute the distance between $\mathbf{m}_i$ and each pixel p in a 2×2 neighbourhood around $\mathbf{m}_i$.

if $D_i < d(p)$, let $d(p) = D_i$ and $L(p) = i$.

3. Update $\mathbf{m}_i$

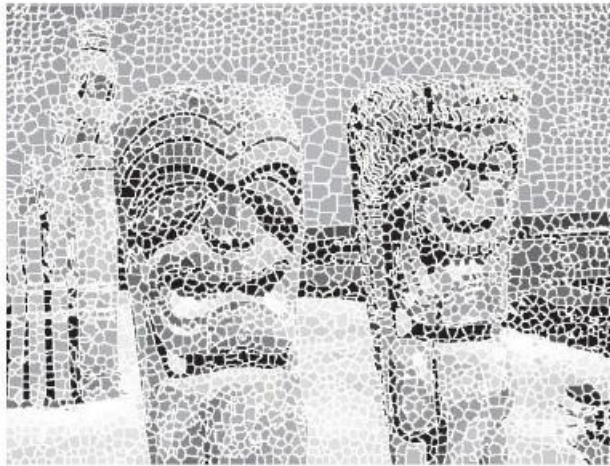
$$\mathbf{m}_i = \frac{1}{|C_i|} \sum_{\mathbf{z} \in C_i} \mathbf{z} \quad i = 1, 2, \dots, n_{sp}$$

4. Like Simple Clustering, Test if the means of the previous steps and the current one are different less than a threshold. If not, repeat step 2. Else, replace all the superpixels by their average value $\mathbf{m}_i$

Superpixels



Original image
(600 x 480 = 480,000 pixels)



Reduced to 4,000 superpixels



40,000 superpixels



1,000 superpixels



500 superpixels



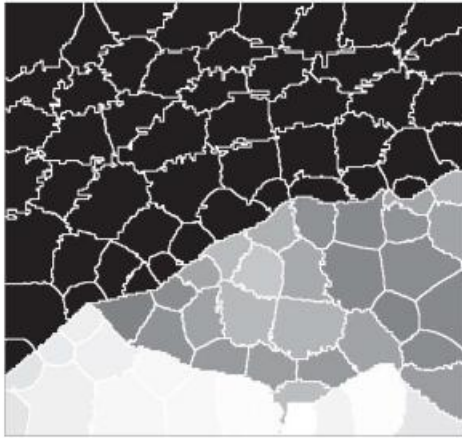
250 superpixels



Original image



Clustered with $k = 3$



100 superpixels



Clustered with $k = 3$ from 100 superpixels



USE OF MOTION IN SEGMENTATION

A Basic Approach

- ▶ One of the simplest approaches is to detect a motion in set of images by detecting change between two image frames $f(x, y, t_i)$ and $f(x, y, t_j)$ taken at time t_i and t_j
- ▶ The two images are compared pixel by pixel and a difference image can be defined as

$$d_{ij}(x, y) = \begin{cases} 1 & \text{if } |f(x, y, t_i) - f(x, y, t_j)| > T \\ 0 & \text{otherwise} \end{cases}$$

where T is a nonnegative threshold

- ▶ This method is frequently used to detect changes in controlled environments such as buildings or parking lots

Accumulative Differences

- ▶ Accumulative difference image (ADI) is formed using a sequence of image frames $f(x, y, t_1), f(x, y, t_2), \dots, f(x, y, t_n)$ in which the values can be calculated as follow

$$A_k(x, y) = \begin{cases} A_{k-1}(x, y) + 1 & \text{if } |R(x, y) - f(x, y, k)| > T \\ A_{k-1}(x, y) & \text{otherwise} \end{cases}$$

$$P_k(x, y) = \begin{cases} P_{k-1}(x, y) + 1 & \text{if } R(x, y) - f(x, y, k) > T \\ P_{k-1}(x, y) & \text{otherwise} \end{cases}$$

$$N_k(x, y) = \begin{cases} N_{k-1}(x, y) + 1 & \text{if } R(x, y) - f(x, y, k) < -T \\ N_{k-1}(x, y) & \text{otherwise} \end{cases}$$

- ▶ Where $A_k(x, y)$, $P_k(x, y)$, and $N_k(x, y)$ are absolute, positive, and negative ADIs respectively.

Accumulative Differences

A rectangle is moving south-east



Absolute ADI



Positive ADI



Negative ADI

Establishing a reference image

- ▶ If we can have one image which contains only stationary objects, this image can be used as a reference image for the future movements
- ▶ However, this kind of image is not always possible. Therefore, we might need to build a reference from a set of images
- ▶ We can obtain information of only-moving objects using positive ADI. Once they move out of the original position, the remains are stationary objects

Establishing a reference image



Car in the middle is moving



Established reference image



ACTIVE CONTOURS: SNAKES AND LEVEL SETS

Background

- ▶ Previous segmentation techniques are based on detecting intensity discontinuities or similarities. But the technique about to be introduced will take approach around the modeling perspective
- ▶ Active contours (or Evolving fronts or Evolving interfaces) are deformable models confined to the plane. “Active” means the contours can be dynamic
- ▶ There are two different paths
 - ▶ Snake
 - ▶ Level sets

Snakes

- ▶ First, we must define a parametric curve in the xy -plane is expressed as

$$(x, y) = (g(s), h(s))$$

- ▶ Where $g(s)$ and $h(s)$ are functions of a parameter s which are called parametric equations.
- ▶ For example, the equation of a circle $x^2 + y^2 = r^2$ can be written in parametric form as

$$\begin{aligned}x &= g(s) = r \cos(s) \\ y &= h(s) = r \sin(s)\end{aligned}$$

Snakes

- ▶ The snake equation is taken in parametric form as follow

$$\begin{aligned}\mathbf{x}(t) &= \mathbf{A}[\mathbf{x}(t-1) + \gamma \mathbf{F}_x(\mathbf{x}(t-1), \mathbf{y}(t-1))] \\ \mathbf{y}(t) &= \mathbf{A}[\mathbf{y}(t-1) + \gamma \mathbf{F}_y(\mathbf{x}(t-1), \mathbf{y}(t-1))]\end{aligned}$$

- ▶ The equation starts at $\mathbf{x}(0)$ and $\mathbf{y}(0)$ and it will be iterated through the image data. The algorithm, stops when $c(t) = c(t-1)$
- ▶ $F(x, y)$ are external forces on the snakes. There are two approaches of obtaining these external forces

External Forces based on The Magnitude of the Image Gradient (MOG)

- ▶ This external force pushes the snake contour toward features of interest in segmentation (edges, etc.)
- ▶ The force components are

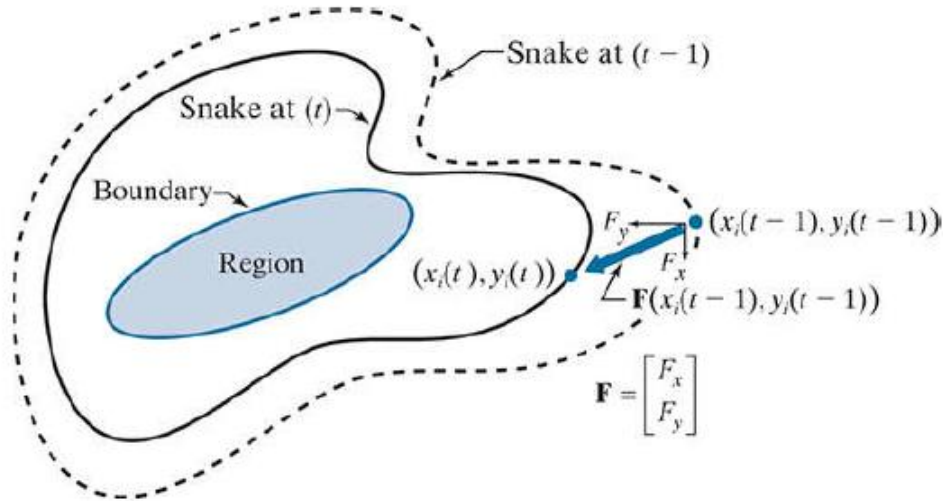
$$F_x(x, y) = F_x(x, y) / (\|\mathbf{F}(x, y)\| + \varepsilon)$$

$$F_y(x, y) = F_y(x, y) / (\|\mathbf{F}(x, y)\| + \varepsilon)$$

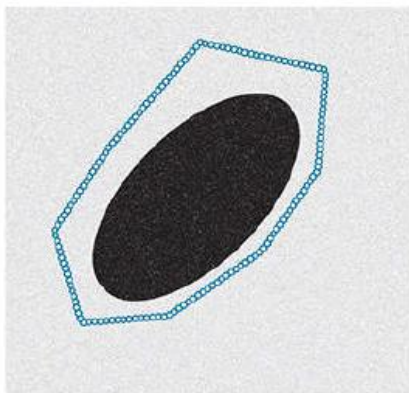
- ▶ Where

$$\|\mathbf{F}(x, y)\| = [F_x(x, y)^2 + F_y(x, y)^2]^{1/2}$$

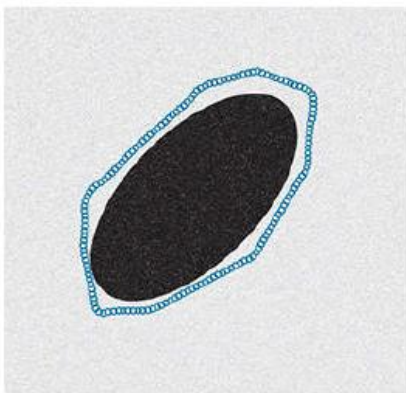
External Forces based on The Magnitude of the Image Gradient (MOG)



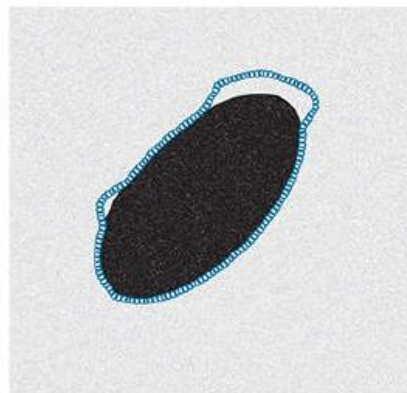
Initial snake



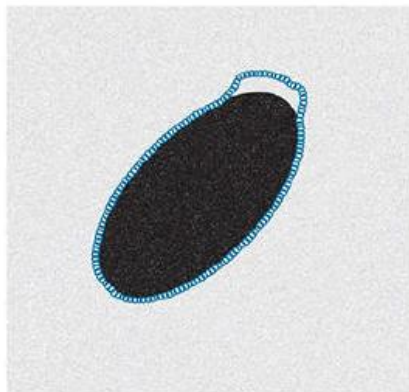
10th iteration



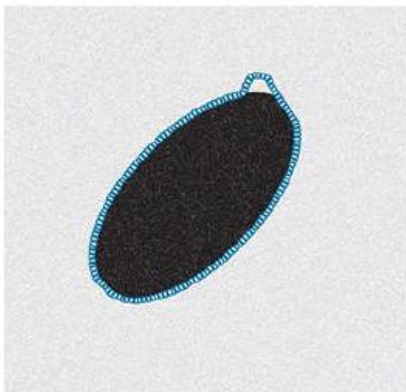
50th iteration



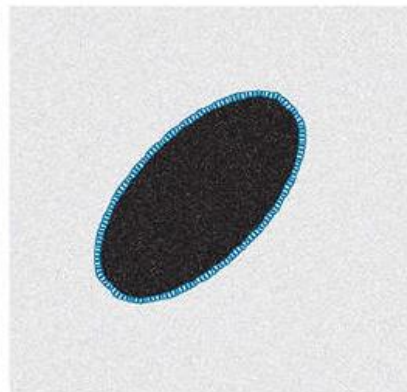
100th iteration



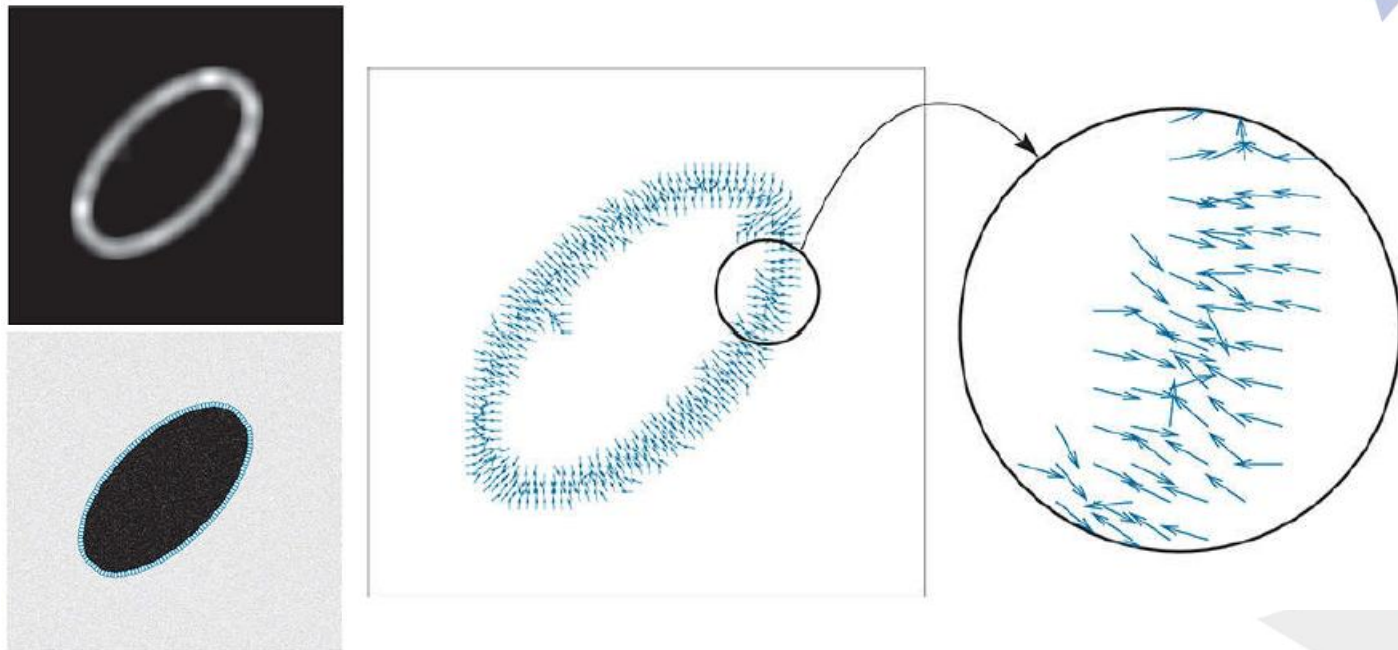
150th iteration



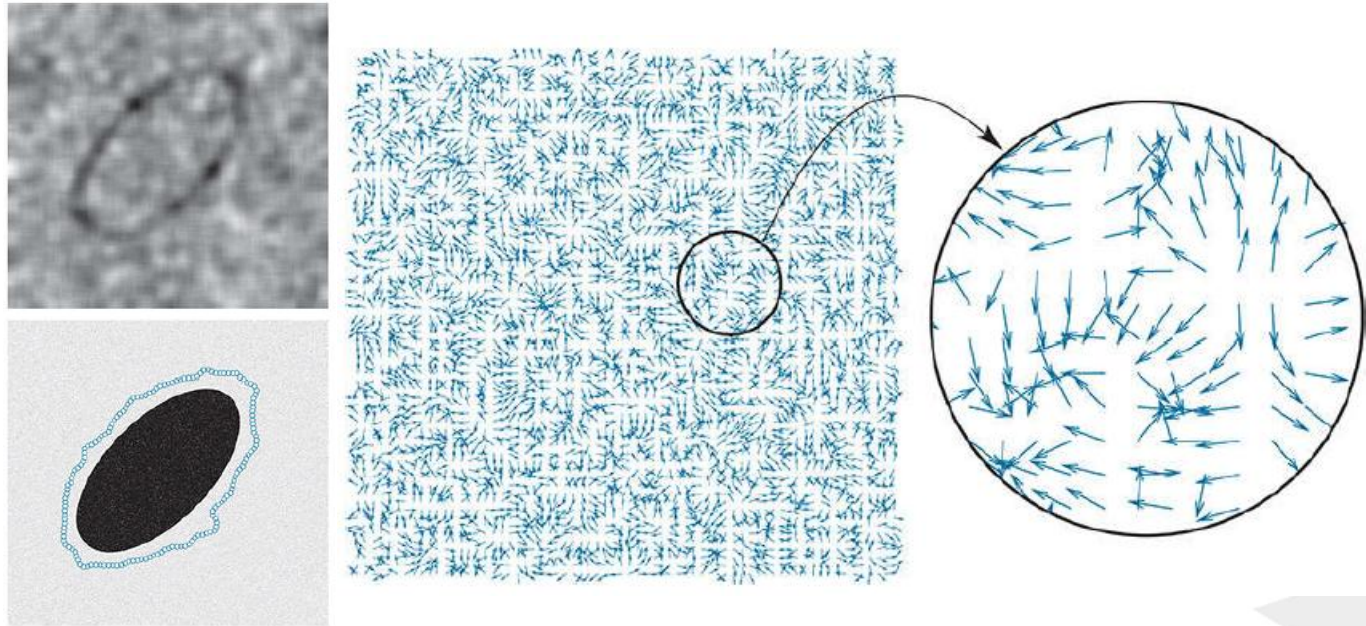
200th iteration



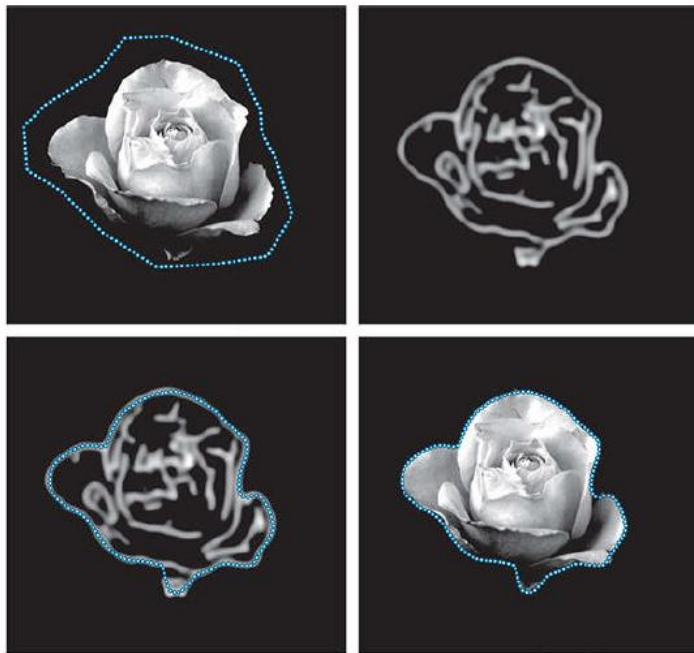
External Forces based on The Magnitude of the Image Gradient (MOG)



External Forces based on The Magnitude of the Image Gradient (MOG)

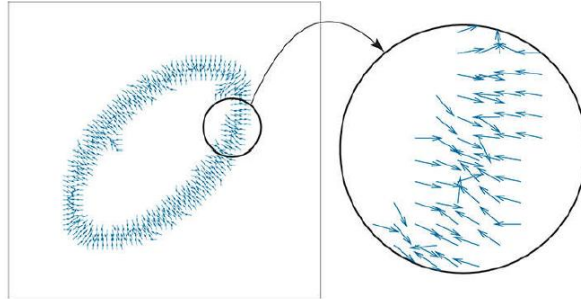


External Forces based on The Magnitude of the Image Gradient (MOG)



External Forces based on Gradient Vector Flow (GVF)

- ▶ External forces from MOG have a setback when the initial snake is too far from the region boundary.



- ▶ Therefore, it is obvious that if we extend force fields further from the boundary, we should have better result.

External Forces based on Gradient Vector Flow (GVF)

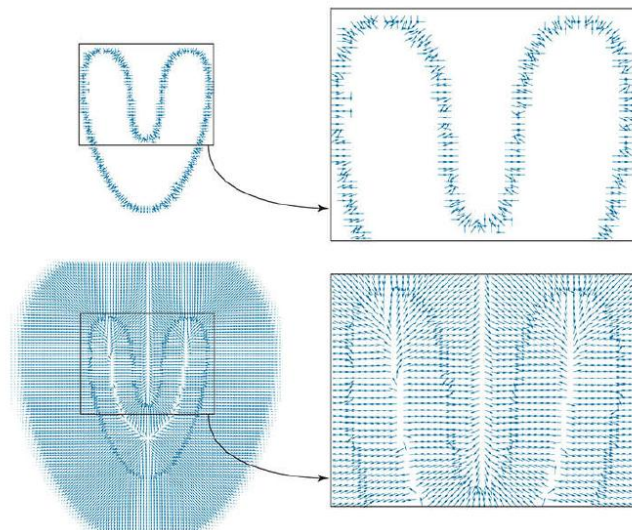
- ▶ The force field of vector flow can be extended using the iterative formula

$$v_x(t) = \mu \nabla^2 v_x(t-1) - \|\nabla e\|^2 \left(v_x(t-1) - \frac{\partial e}{\partial x} \right)$$

$$v_y(t) = \mu \nabla^2 v_y(t-1) - \|\nabla e\|^2 \left(v_y(t-1) - \frac{\partial e}{\partial y} \right)$$

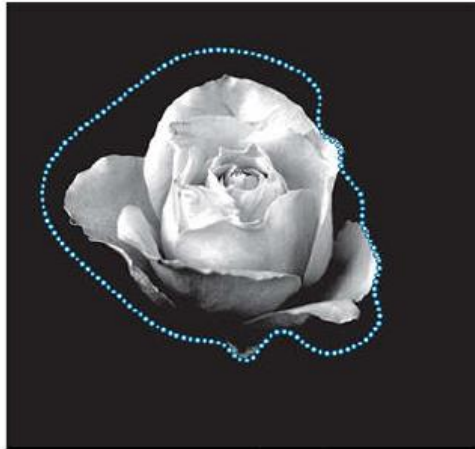
External Forces based on Gradient Vector Flow (GVF)

- An MOG field can be extended to a GVF field as follow



External Forces based on Gradient Vector Flow (GVF)

- ▶ GVF is generally faster to find the boundary than MOG in the same number of iterations



MOG 90th iteration



GVF 90th iteration

Level Sets

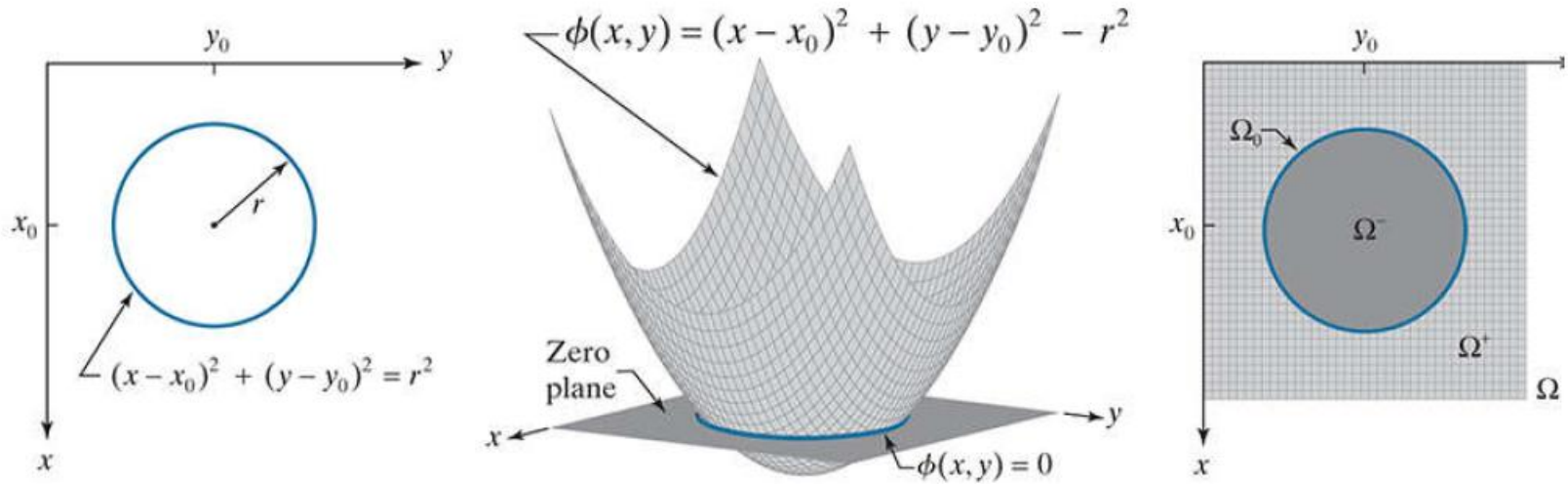
- ▶ Level sets are sets of points of a 2-D curve formed by the intersection of a plane and a 3-D surface
- ▶ Using this technique, it can start with a small region and be able to adapt to a changing topology
- ▶ We can start with the equation of a circle.

$$(x - x_0)^2 + (y - y_0)^2 - r^2 = 0$$

- ▶ Thus, we can expand the equation to the 3-D space with a scalar function

$$\phi(x, y) = (x - x_0)^2 + (y - y_0)^2 - r^2$$

Level Sets

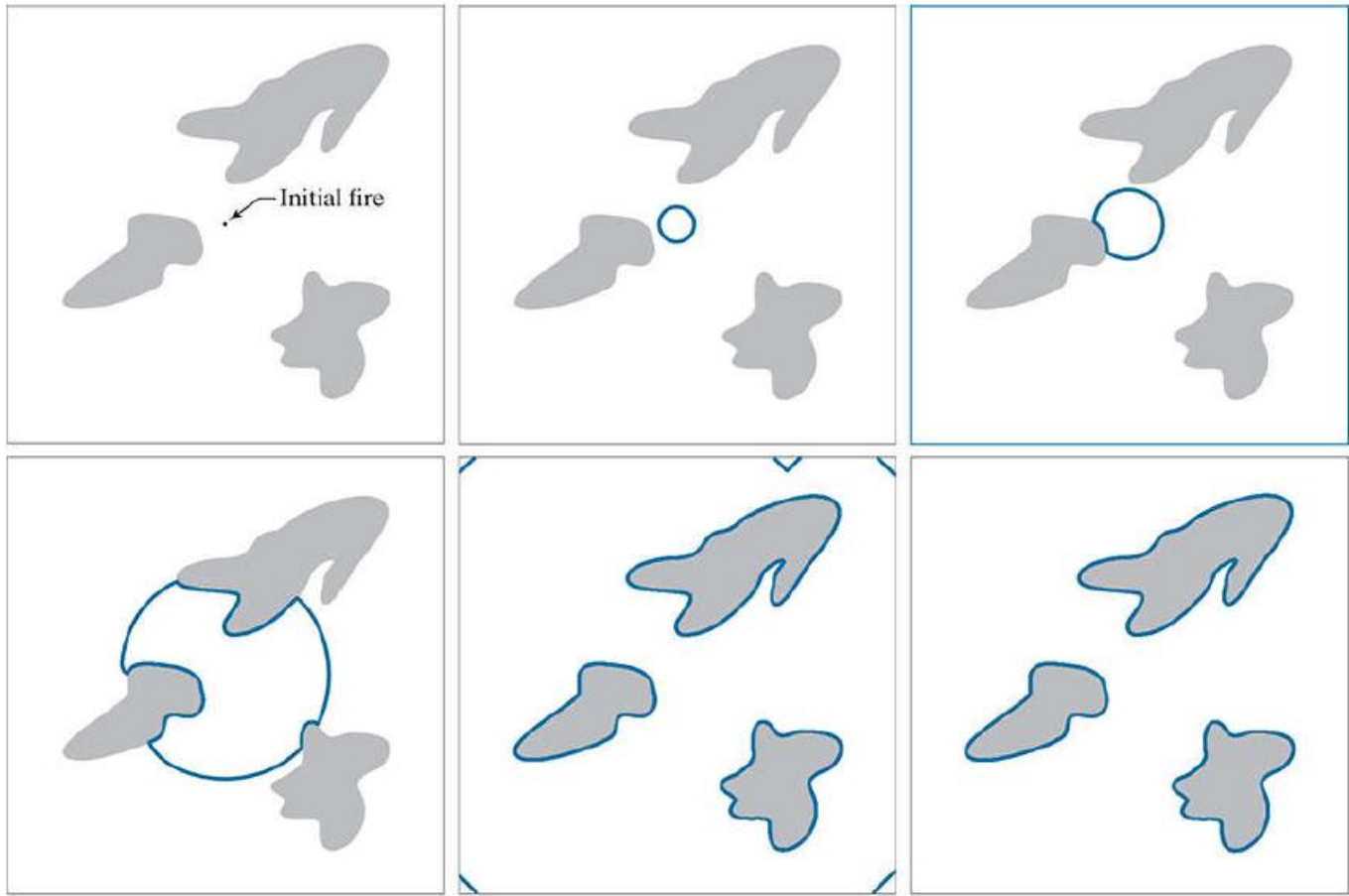


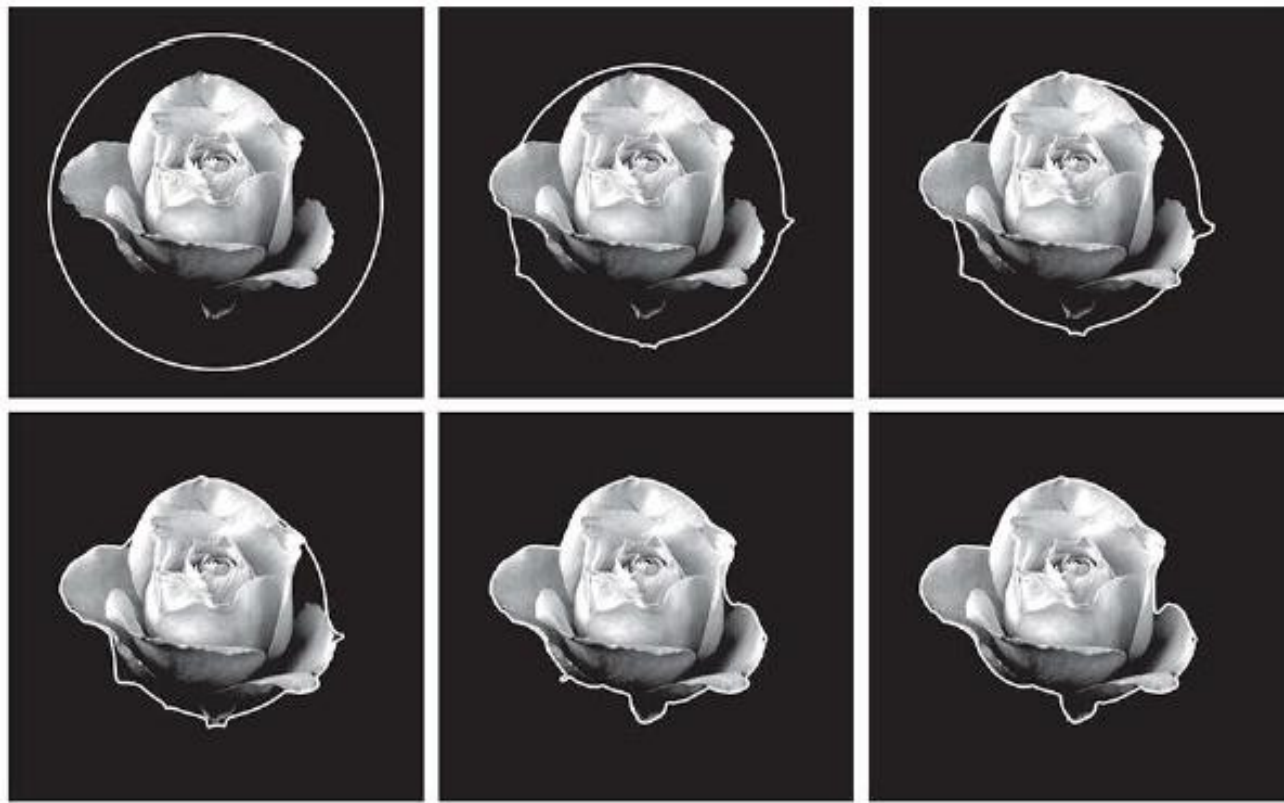
Level Sets

- ▶ The iterative equation of 2-D upwind norms of level sets has the form as

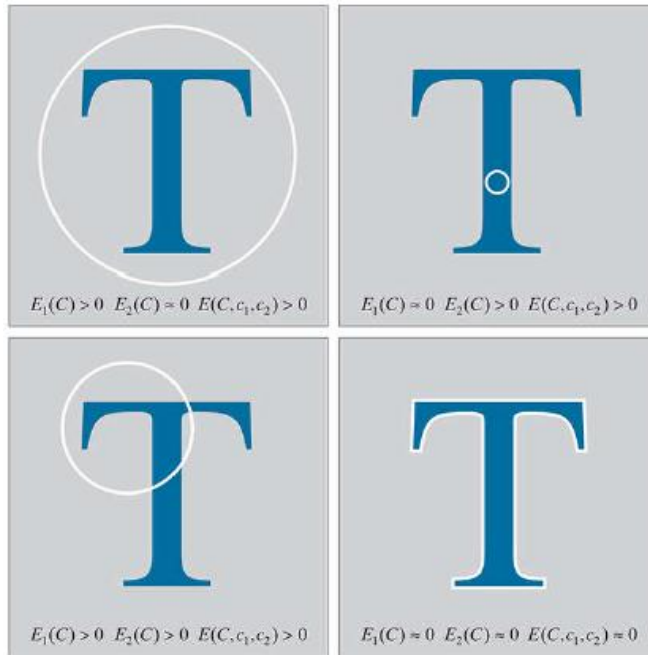
$$\phi^{n+1} = \phi^n - \Delta t \{ \max(F, 0) \|\nabla \phi^n\|^+ + \min(F, 0) \|\nabla \phi^n\|^- \}$$

- ▶ Where $\phi(x, y, t) = \sqrt{x^2 + y^2} - r - t$
- ▶ The iteration stops when $\phi^{n+1} = \phi^n$ within a stopping tolerance bound (a number of defined iterations)

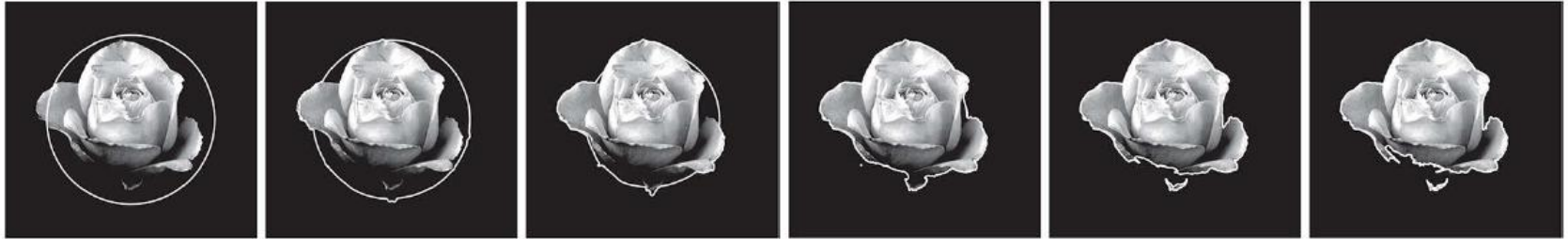
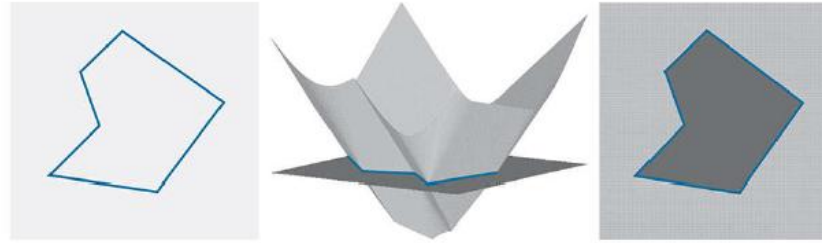




Advanced Level Sets



Advanced Level Sets



Advanced Level Sets

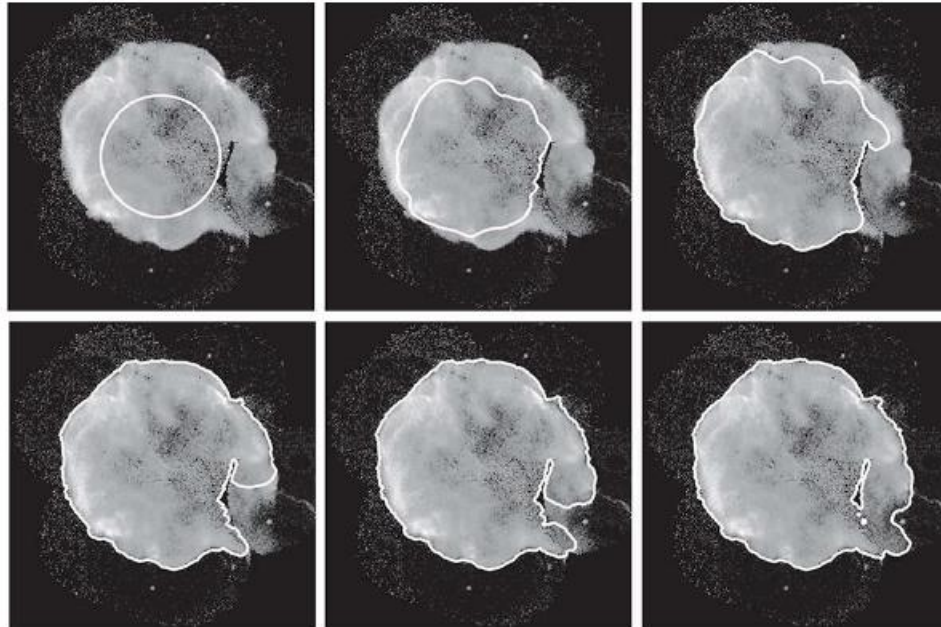


Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.



End of Week 11